MODUL PRAKTIKUM 7

Machine Learning untuk Prediksi Traffic
(*Smart City AI*)

Programming Language	: **Python**
Development Environment	: ***Linux Server Environment***
Code Editor	: **VS Code** (*Remote* ***WSL***)
Command Line Interface	: **Bash**
Distributed Data Processing Engine	: **Apache Spark (PySpark)**
Machine Learning Models	: **Random Forest**
Real-Time Visualization	: **Streamlit**
Durasi	: **150 – 180 menit**

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1. TUJUAN PRAKTIKUM

Mahasiswa mampu:

1. Mengimplementasikan **Machine Learning untuk prediksi traffic**
2. Menggunakan pipeline data berbasis project Big Data
3. Melakukan **feature engineering time-series**
4. Membuat **dashboard interaktif (Streamlit modern UI)**
5. Mengintegrasikan hasil ke dalam struktur project

2. KONTEKS INDUSTRI

Dalam Smart City:

- Adaptive Traffic Control System
- Prediksi kemacetan real-time
- Smart routing (Google Maps, Waze)
- Integrasi IoT sensor jalan

Perusahaan yang menerapkan:

- Google (Traffic Prediction AI)
- IBM Smart City
- Siemens Mobility

3. ARSITEKTUR PIPELINE

Pipeline Praktikum 7

```
data/raw → data/clean → analytics → model → dashboard → insight
```

3.1 Konsep yang Digunakan

Konsep	Penjelasan
Time Series	Data berbasis waktu (jam/hari)
Supervised Learning	Model belajar dari data historis
Regression	Prediksi jumlah kendaraan
Feature Engineering	Jam, hari, lag
Model ML	Random Forest
Evaluation	MAE

4. LANGKAH-LANGKAH PRAKTIKUM

STEP 1 — Tambahkan Dataset (RAW)

👉 Download dataset: traffic_smartcity_v1.csv (https://muhayat-lab.github.io/courses/big-data-technology)

📁 Simpan file dataset tersebut pada folder project:

bigdata-project/data/raw/traffic_smartcity_v1.csv

STEP 2 — Data Cleaning

📁 Buat file baru:

scripts/traffic_data_cleaning_v1.py

```
import pandas as pd

df = pd.read_csv('data/raw/traffic_smartcity_v1.csv')

df['datetime'] = pd.to_datetime(df['datetime'])
df = df.sort_values('datetime')
df = df.dropna()

df.to_csv('data/clean/traffic_smartcity_clean_v1.csv', index=False)

print("Data cleaning selesai")
```

STEP 3 — Feature Engineering + Modeling

📁 Buat file baru:

analytics/traffic_ml_model_v1.py

```
import pandas as pd
from sklearn.ensemble import RandomForestRegressor
import joblib
```

```
df = pd.read_csv('data/clean/traffic_smartcity_clean_v1.csv')

df['datetime'] = pd.to_datetime(df['datetime'])
df['hour'] = df['datetime'].dt.hour
df['day'] = df['datetime'].dt.dayofweek

df['lag1'] = df['traffic'].shift(1)
df = df.dropna()

X = df[['hour','day','lag1']]
y = df['traffic']

model = RandomForestRegressor()
model.fit(X, y)

joblib.dump(model, 'models/traffic_model_v1.pkl')

print("Model berhasil disimpan")
```

📁 Tambahkan **folder baru:**

`models/`

STEP 4 — Dashboard Streamlit (Modern UI)

📁 Buat file baru:

`dashboard/traffic_dashboard_v1.py`

```
import streamlit as st
import pandas as pd
import joblib
import matplotlib.pyplot as plt

st.set_page_config(page_title="Smart Traffic AI", layout="wide")

st.title("🚦 Smart City Traffic Dashboard")

# Load data
df = pd.read_csv('data/clean/traffic_smartcity_clean_v1.csv')

# Load model
model = joblib.load('models/traffic_model_v1.pkl')

# Feature
df['datetime'] = pd.to_datetime(df['datetime'])
df['hour'] = df['datetime'].dt.hour
df['day'] = df['datetime'].dt.dayofweek
df['lag1'] = df['traffic'].shift(1)
df = df.dropna()
```

```
# Metrics
col1, col2 = st.columns(2)

col1.metric("Avg Traffic", int(df['traffic'].mean()))
col2.metric("Max Traffic", int(df['traffic'].max()))

# Chart
st.subheader("📈 Traffic Trend")

fig, ax = plt.subplots()
ax.plot(df['traffic'].values)
st.pyplot(fig)

# Prediction
st.subheader("🔮 Prediksi Traffic")

hour = st.slider("Jam", 0, 23, 17)
day = st.slider("Hari", 0, 6, 2)
lag1 = st.number_input("Traffic sebelumnya", 50, 300, 120)

if st.button("Prediksi"):
    pred = model.predict([[hour, day, lag1]])
    st.success(f"Prediksi: {int(pred[0])} kendaraan")
```

▶ STEP 5 — Jalankan Data Cleaning – ML Model - Dashboard

1. Jalankan Data Cleaning pada terminal 1:

   `python scripts/traffic_data_cleaning_v1.py`

2. Jalankan ML Model pada terminal 2:

   `python analytics/traffic_ml_model_v1.py`

3. Jalankan Dashboard pada terminal 3:

   `streamlit run dashboard/traffic_dashboard_v1.py`

5. CHECKLIST VALIDASI PRAKTIKUM

No	Validasi	Status
1	**Dataset raw tersedia**	☐
2	**Data clean berhasil**	☐
3	**Model tersimpan (.pkl)**	☐
4	**Dashboard berjalan**	☐
5	**Prediksi muncul**	☐
6	**Tidak error path**	☐

6. OUTPUT WAJIB YANG DIKUMPULKAN

Mahasiswa wajib submit:

1. 📁 Struktur project (screenshot)
2. 📄 Script:
   - cleaning
   - modeling
   - dashboard
3. 📊 Screenshot dashboard
4. 📈 Nilai prediksi
5. 📝 Laporan (max 3 halaman)

7. RUBRIK PENILAIAN (SKOR 1–4)

Aspek	1	2	3	4
Struktur project	Tidak sesuai	Kurang	Sesuai	Sangat rapi
Data pipeline	Tidak jalan	Sebagian	Jalan	Optimal
Model ML	Tidak ada	Sederhana	Baik	Optimal
Dashboard	Error	Minimal	Baik	Profesional
Insight	Tidak ada	Dangkal	Baik	Mendalam

8. INSIGHT

🎯 **Insight Utama:**

- Machine Learning sangat powerful untuk prediksi traffic
- Feature sederhana (jam, hari) sudah cukup kuat
- Pipeline data penting dalam Big Data system

💡 **Insight Lanjutan:**

- Integrasi real-time → Kafka / Spark
- Model lanjutan → LSTM
- Smart City → AI-driven decision system

✅ KESIMPULAN

Dengan praktikum ini, kita telah membangun:

✓ Pipeline Big Data sederhana
✓ Model Machine Learning
✓ Dashboard profesional
✓ Simulasi Smart City AI